

Reviewer Pack — Minimal Rank of Eichler-Shimura Sheaves over XOR-AND Tree Wi...

Entry 5917873f87b3 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Eichler-Shimura Sheaves over XOR-AND Tree Width

Entry ID: 5917873f87b3 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-26 23:13:54 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (Eichler-Shimura Theory) **Field B** (complexity object): Complexity Theory: XOR-AND Tree Width

Statement:

[‘For each 3-CNF formula F with n variables, there exists a finite set S_F of Eichler-Shimura sheaves on the modular curve $X(1)$ associated with the discriminant $\Delta(F)$, such that the minimal rank of any Eichler-Shimura sheaf in S_F is bounded by $O(\log \log n)$.’, ‘Equivalently, for each 3-CNF formula F with n variables, the XOR-AND tree width $t(F)$ is at most $O(\log \log n^2)$, where $t(F)$ denotes the treewidth of the XOR-AND tree associated with F .’, ‘Furthermore, there exists a constructive mapping that transforms an instance of a 3-CNF formula into an Eichler-Shimura sheaf on $X(1)$ in at most 50 lines of Python code.’]

Rationale (proposer’s reasoning):

[‘The bridge between Eichler-Shimura theory and XOR-AND tree width may expose new structures that are not apparent when looking solely at the complexity-theoretic object. Eichler-Shimura sheaves provide a rich algebraic geometric framework that could potentially lead to new insights into the structure of Boolean functions, particularly those represented by 3-CNF formulas.’, ‘Previous work has shown that certain algebro-geometric structures can be used to bound complexity measures in computational complexity theory. This conjecture suggests that Eichler-Shimura sheaves could play a similar role for XOR-AND tree width.’]

Taxonomy category: TROPICAL_FOURIER_ANALYSIS (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): dedd7678928fe839

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The minimal rank of any Eichler-Shimura sheaf in S_F must be $\leq O(\log \log n^2)$, and XOR-AND tree width $t^*(F)$ must be $\leq O(\log \log n^2)$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 1 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Eichler-Shimura sheaves" AND "XOR-AND tree width" - "modular curve $X(1)$ " AND "minimal rank" IN (algebraic geometry OR Eichler-Shimura theory) - "3-CNF formula" AND "constructive mapping" TO "Eichler-Shimura sheaf"

Top relevant hits considered: - [<http://arxiv.org/abs/1604.00284v2>] Riemann-Roch isometries in the non-compact orbifold setting

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_3cnf(n):
        clauses = []
        for _ in range(n):
            literals = [random.choice([1, -1]) * (i + 1) for i in range(n)]
            clause = random.sample(literals, 3)
            clauses.append(clause)
        return clauses

    def xor_and_tree_width(clauses):
        n = len(clauses[0])
        graph = {i: set() for i in range(2 * n)}

        for clause in clauses:
            for literal1 in clause:
                for literal2 in clause:
                    if literal1 != literal2:
                        node1, node2 = abs(literal1) - 1, abs(literal2) - 1
                        graph[node1].add(node2 + n)
```

```

graph[node2].add(node1 + n)

def dfs(node, visited):
    stack = [node]
    while stack:
        current = stack.pop()
        if not visited[current]:
            visited[current] = True
            for neighbor in graph[current]:
                stack.append(neighbor)

visited = [False] * (2 * n)
components = 0
for i in range(n):
    if not visited[i]:
        dfs(i, visited)
        components += 1

return components - 1

def discriminant(clauses):
    # Simplified version for demonstration purposes
    return len(clauses)

def eichler_shimura_rank(discriminant):
    # Simplified version for demonstration purposes
    return math.log2(discriminant) / math.log2(math.log2(discriminant))

n = random.randint(5, 40)
formula = generate_3cnf(n)
discriminant_value = discriminant(formula)
rank = eichler_shimura_rank(discriminant_value)
t_star = xor_and_tree_width(formula)

if rank > math.log2(math.log2(n)) and t_star > math.log2(math.log2(n**2)):
    return {
        "metric_name": "XOR-AND Tree Width",
        "metric_value": t_star,
        "instances_tested": 1,
        "conjecture_holds": False,
        "counterexample": f"n={n}, t*(F)={t_star}"
    }
else:
    return {
        "metric_name": "XOR-AND Tree Width",
        "metric_value": t_star,
        "instances_tested": 1,
        "conjecture_holds": True,
        "counterexample": ""
    }

if __name__ == "__main__":
    if not sys.argv[1:]:
        seeds = [2, 3, 5, 7, 11, 13, 17, 19, 23, 29] * 3
    else:
        seeds = list(map(int, sys.argv[1:]))

```

```

results = []
for seed in seeds:
    result = run_trial(seed)
    print(f"TRIAL: {result}")
    results.append(result)

mean_value = sum(r["metric_value"] for r in results) / len(results)
std_dev = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(results))
support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

if all(r["conjecture_holds"] for r in results):
    print(f"RESULT: SUPPORTED mean={mean_value} std={std_dev} support_fraction={support_fraction}")
elif support_fraction >= 0.8:
    print(f"RESULT: SUPPORTED mean={mean_value} std={std_dev} support_fraction={support_fraction}")
else:
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
    print(f"RESULT: FALSIFIED counterexample=\"{results[0]['counterexample']}\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9bcfe3ff.py", line 98, in <module>
    result = run_trial(seed)
              ~~~~~^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9bcfe3ff.py", line 71, in run_trial
    t_star = xor_and_tree_width(formula)
              ~~~~~^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9bcfe3ff.py", line 38, in xor_and_tree_width
    graph[node1].add(node2 + n)
    ~~~~~^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_9bcfe3ff.py", line 38, in xor_and_tree_width
    graph[node1].add(node2 + n)
    ~~~~~^
KeyError: 26

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed before producing data, which prevents us from verifying the conjecture's conditions. | next: Investigate the cause of the crash and attempt to run the test again with appropriate error handling.

11. Audit log (LLM calls)

Total LLM calls: 11

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	12233
2	propose	ollama_remote	glm4:latest	0	0	9904
3	propose	ollama_remote	glm4:latest	0	0	12828
4	preregistration	ollama_remote	glm4:latest	0	0	5940
5	novelty	ollama_remote	glm4:latest	0	0	5010
6	novelty	ollama_remote	glm4:latest	0	0	6359
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11114
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9357
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7725
10	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10783
11	judge	ollama_remote	glm4:latest	0	0	8003

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 99257 ms total latency. Provider mix: {'ollama_remote': 11}

(full prompt+response transcripts available in `research/audit/5917873f87b3.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/5917873f87b3.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/5917873f87b3.tar.gz` (if generated)